Travel Medicine and Infectious Disease (2007) 5, 230–235

Available at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevierhealth.com/journals/tmid

Medical treatment at Louis Armstrong New Orleans International Airport after hurricane Katrina: The experience of disaster medical assistance teams WA-1 and OR-2

Christopher Sanford[a,*], Jonathan Jui[b,c], Helen C. Miller[c], Kathleen A. Jobe[d]

[a]Department of Family Medicine, School of Medicine, University of Washington, Seattle, WA, USA
[b]Emergency Medicine, Oregon Health & Science University, Portland, OR, USA
[c]Emergency Medical Services, Multnomah County Health Department, Portland, OR, USA
[d]Division of Emergency Medicine, University of Washington, Seattle, WA, USA

Received 2 March 2007; received in revised form 8 March 2007; accepted 14 March 2007
Available online 26 April 2007

KEYWORDS
Disaster medicine;
Complex humanitarian emergency;
Hurricane;
Flood;
Katrina;
NDMS (National Disaster Medical System)

Summary
In the week following Hurricane Katrina, over 3000 patients were evacuated by air from a triage and medical treatment station at the Louis Armstrong New Orleans International Airport. This represents the largest air evacuation in history. Over 24,000 additional evacuees were transported from the airport to shelters. Disaster Medical Assistance Teams (DMATs) from several US states were deployed to the Louis Armstrong New Orleans International Airport to provide medical care to those evacuated from New Orleans. Despite warning from the US National Weather Service of catastrophic damage to New Orleans, adequate medical staffing was not attained at the airport triage station until 6 days after the hurricane struck. Organizational lapses, including inadequate medical and operational planning, understaffing of medical personnel, and failure to utilize Incident Command System, diminished the effectiveness of the Hurricane Katrina New Orleans Medical Operation.
© 2007 Elsevier Ltd. All rights reserved.

Introduction

At dawn on Monday, August 29, 2005, Hurricane Katrina, a Category 3 hurricane with sustained winds of 179–209 km/h,[1] made landfall at the Gulf Coast of the United States. Traveling northward along the Louisiana–Mississippi border, the hurricane passed 32 km to the east of New Orleans,

*Corresponding author. Tel.: +1 206 685 1060; fax: +1 206 221 5368.
E-mail addresses: casanfo@u.washington.edu (C. Sanford), jui@ohsu.edu (J. Jui), millerhelen@comcast.net (H.C. Miller), kaj@u.washington.edu (K.A. Jobe).

1477-8939/$ - see front matter © 2007 Elsevier Ltd. All rights reserved.
doi:10.1016/j.tmaid.2007.03.002

Louisiana, USA, population 500,000. While essentially destroying the cities of Gulfport and Biloxi, Mississippi, the initial impression was that New Orleans had escaped the brunt of the storm. However, the following day, levees along two blocks of the 17th Street Canal and at the Industrial Canal failed, and New Orleans, most of which is situated below sea level, was flooded. By the following day, 80% of New Orleans was under water at depth of up to 6 m.

While a majority of New Orleans residents followed New Orleans Mayor C. Ray Nagin's order to evacuate on Saturday, 2 days before the hurricane making landfall, 80,000–100,000 residents remained in New Orleans, including a large population of hospital patients and nursing home residents. Over 1300 deaths attributable to Hurricane Katrina have been confirmed, making this the worst natural disaster on US soil since a hurricane struck Galveston, Texas, in 1900, killing 6000–12,000 people. The evacuation of New Orleans represents the first evacuation of a major American city since Richmond and Atlanta during the Civil War (1861–1865).

Background: formation of DMATs

Disaster Medical Assistance Teams (DMATs) date back to 1984, when the United States Public Health Service (USPHS) established two DMAT prototypes.[2] Comprised of volunteer healthcare professionals, team members can be federalized and deployed by National Disaster Medical System (NDMS) to disasters in the United States and its territories. There are currently 61 DMATs in the United States[3]; approximately half of these are Level 1 (fully deployable within 8 h) at this time.

One author of this paper [Jobe] served as a medical officer on DMAT WA-1; the other three [Miller, Jui, and Sanford] served on DMAT OR-2, as, respectively, team commander, chief medical officer, and medical officer. Both WA-1 and OR-2 are Level 1 teams. DMAT WA-1 was initially formed as a private team in 1985 as the Seattle-King County Disaster Medical Team, which responds to international disasters and the needs of developing nations. It subsequently became a DMAT in 1993. DMAT OR-2 was established in 1999.

Each deployed DMAT is comprised of approximately 35 team members: 3–4 physicians, several mid-level practitioners (physicians' assistants and nurse practitioners), several nurses, and pharmacy, communications, logistics, and safety staff.

The Louis Armstrong New Orleans International Airport

The Louis Armstrong New Orleans International Airport is located 24 km northwest of downtown New Orleans. The airport has six runways, 42 passenger boarding gates and ordinarily sees approximately 4300 flights per month.[4,5] It is situated above sea level, and while it sustained significant damage from wind and rain during Hurricane Katrina, it did not flood. Following Hurricane Katrina, the airport was designated to be the casualty collection facility, as well as the field triage and treatment center for medical evacuees from New Orleans and surrounding areas.

During the days following Hurricane Katrina, the upper and lower levels of Concourse D served as the primary medical staging area. This area was chosen because it was the least storm-damaged portion of the airport complex, and was able to accommodate large aircraft for NDMS and rescue operations. An additional feature of this concourse was its large skylight which offered light during daytime during a period when commercial electricity was not available. A bar on Concourse D was converted into a pharmacy. A corner of the concourse adjacent to the pharmacy served as the logistics depot for the DMATs' materiél.

The experiences of DMATs WA-1 and OR-2

One of the authors [Jobe] flew to Houston, Texas, on Sunday, August 28 as a Medical Officer on DMAT WA-1. DMATs from New Mexico, California and Oklahoma were also deployed to Houston on the same day. The following day DMAT WA-1 drove in convoy to Baton Rouge, which is situated approximately 97 km northwest of New Orleans. DMAT WA-1 was then directed to drive to New Orleans' Superdome, a covered sports stadium with a capacity of 69,000 during sports events. By the time the convoy of vehicles reached the Superdome at approximately 4:30 a.m. on Tuesday, August 30, floodwater reached the floorboards of vehicles and was rising. Due to worsening flooding and deteriorating security conditions inside the Superdome, DMAT WA-1 was directed to drive back to Baton Rouge. DMAT WA-1 returned to Baton Rouge to await further instructions from NDMS. After waiting at Baton Rouge for a few hours, they were directed to drive to the New Orleans International Airport, which they reached at approximately 1 a.m. on Wednesday, August 31. Two other DMATs also arrived at the airport at approximately the same time: CA-4 and TX-4. These teams' mission was to provide triage and medical care to an anticipated 2500 patients.

Medical triage was performed as per the schema summarized by Briggs and Leong.[6] At the triage station in the airport, those who did not require medical attention were sent to an area where they waited for transport to a shelter. Those requiring medical assistance were sent to one of four areas: "Green Tent," which saw patients with minor problems, "Yellow Tent," which cared for those with significant but not immediately life-threatening medical problems, or "Red Tent," which saw people with life-threatening medical problems. Those believed to be likely to die within 24 h despite medical intervention were sent to an area designated as an expectant area ("Black Tent"), in which they received nursing care but no medical interventions on the model of hospice care.

Within 15 h of the arrival of three DMATs at the airport, the number of evacuees at the airport increased to several thousands, and providers were overwhelmed. Patients arrived by helicopter, ambulance and bus from hospitals, nursing homes, search and rescue operations, and emergency medical services; some self-evacuated to the airport. DMAT staff worked around the clock but was not able to offer basic nursing care, let alone medical care, to the majority of the ill and injured. Medical staff grew exhausted but continued to work as supplies ran low and evacuees

continued to stream into the airport. Small electrical generators at the airport provided sufficient electric power for a few lights but not for air conditioning. There was no potable running water, and mass care, feeding and sanitation emergency service functions were absent. The indoor temperature was an estimated 38 °C (100 F), with high humidity. The triage station had three entrances, which exacerbated problems with crowd control. Security screening of evacuees and patients was minimal or absent. On-site DMATs stated that they had made repeated requests for additional DMAT personnel and supplies.

DMAT OR-2 was activated on Tuesday, August 30, and arrived at the New Orleans International Airport at 15:00 h on Thursday, September 1. Three of the authors of this paper (Sanford, Jui, and Miller) arrived with DMAT OR-2. The situation at the airport when DMAT OR-2 arrived was one of minimally contained chaos. Approximately 500 patients lay on the floor, many gravely ill. Thousands of evacuees, some ill, some merely in need to triage to shelters, milled in adjacent halls. The most basic needs of the ill—water, food, and access to toilets or bedpans—were not being met.

After an initial tour through the airport, DMAT OR-2 personnel were stunned; many were in tears. DMAT members and on-site security staff, who included Vietnam War veterans and medical responders to the terrorist attack on the World Trade Center, September 11, 2001, uniformly stated that the situation at the New Orleans airport in the wake of Hurricane Katrina was "The worst thing I've ever seen."

A physician on a Texas DMAT at the airport stated, "All we could do was provide the barest amount of comfort care. We watched many, many people die. We practiced medical triage at its most basic, black-tagging the sickest people and culling them from the masses so that they could die in a separate area. We were so short on wheelchairs and litters that we had to stack patients in airport chairs and lay them on the floor. They remained there for hours, too tired to be frightened, too weak to be cared about their urine- and stool-soaked clothing, too desperate to even ask what was going to happen next".[7]

A minority of patients seen at the airport was injured by the direct effects of the hurricane; the majority was acutely ill as the result of the abrupt withdrawal of medical care from a chronically ill population. Providers cared for insulin-dependent diabetics who had not had insulin for 5–7 days, chronic renal failure patients who had not had dialysis for 5–7 days, hypertensives who had not taken their antihypertensive medications for several days who suffered from strokes and myocardial infarctions even as they were being evaluated. The acutely ill and injured included those with severe head injuries, pulmonary edema, sepsis, peritonitis, and respiratory failure. Patients included those transported directly from intensive care units, some of whom had undergone major surgery, including brain surgery and organ transplant surgery, only days before. While a minority of patients was accompanied by their medical records, a majority was not.

DMAT members worked the flightline to receive evacuees and triage into patient and non-medical groups. DMAT members transported stretcher patients from arriving aircraft into the baggage level of Terminal D. DMATs were assisted by the military and US Forest Service with this task starting on the third day.

DMAT members slept in the baggage claim area, on the luggage belts directly or adjacent to luggage carousels when cots were available. Frequent loud overhead announcements, barking dogs, passers-by, and security concerns, represented significant barriers to sleep.

In light of the civil unrest at the Superdome in New Orleans, and the large, potentially volatile crowds present at the airport, an emergency evacuation plan was devised by DMAT Safety Officers and relayed to all DMAT members. Direct violence by evacuees against DMAT healthcare responders was rare, but not absent.

Twenty-three of New Orleans' 26 hospitals were flooded or otherwise incapacitated by Hurricane Katrina; virtually all of these patients were transported to the airport. Similarly, the residents of a number of nursing homes and convalescent hospitals were emptied into the airport. Helicopters, buses, and ambulances delivered those rescued from rooftops and overpasses to the airport.

Lt. Gen. Michael W. Wooley, Air Force Special Operations Command (AFSOC) commander, stated that approximately 5500 evacuees were plucked from rooftops by helicopters, pulled into rescue Zodiac boats, or otherwise evacuated to safety; virtually 100% of these people were processed through the airport triage and treatment station.[8] Chris Trevino, MD, Medical Director, Emergency Medical Services, State of Louisiana, mobilized a vast fleet of ambulances and buses that were utilized to transport patients and evacuees to hospitals and shelters in Baton Rouge and elsewhere.

The air traffic at the New Orleans International Airport following Hurricane Katrina was phenomenal. A C-130 pilot estimated that he saw a helicopter land "every 15–45 s" while he was at the airport.[9] An Air Force Medical Service website put the frequency of arriving helicopters at one every 10–15 s.[10] These estimates are corroborated by a Federal Aviation and Aeronautics (FAA) fact sheet that states that during peak times, up to 150 helicopters per hour landed and took off.[11]

The Journal of the [US] Air Force website says that as of September 4, Air Force aircraft had transported over 2955 patients from the airport.[12] "During the next critical days, the Air National Guard flew more than 300 sorties a day, said Lt. Gen. Daniel James III, ANG director."[13] Over labor day weekend [September 3–5] the airport handled 3300 flights per day.[14]

On the afternoon of Thursday, September 1, the USFS Type 1 National Overhead team (Red Team) responded to the airport as part of the national response plan. This team's mission was to provide management and logistical support to all personnel working at the airport. Within hours, this team provided water, food, and handwashing stations for personnel. Within 48 h showers were made available to airport staff including medical personnel. The US Forest Service also provided key communications infrastructure to unite all providers (LE, Medical, Overhead, and Airport, and military) on similar channels.

Although disaster planners never intended the New Orleans International Airport to be a staging facility for refugees or a definitive medical center, of necessity it took on those roles in the weeks following Hurricane Katrina. At least three babies, including a set of twins, were born at the

airport in the week following Hurricane Katrina. The peak airport population of patients and evacuees were estimated to be 16,000 people.[15] The actual number of evacuees triaged, treated, and transported are unknown but exceed official tallies; in the initial days following the hurricane record keeping was at best intermittent. Numerous pets, primarily dogs and cats, were evacuated to the airport, many of which received medical care from a Veterinary Medical Assistance Teams (VMATs), which arrived on September 4th.

On Saturday, September 3rd, the arrival of additional staff and air evacuation resources allowed re-triage of survivors of the hospice to hospitals for in-patient care. The combination of the arrival of additional DMATS, the evacuation of patients, and a markedly decreased incoming patient flow, resulted in teams already on-site transitioning from being overworked to being under-utilized within the span of a few hours. Patient loads remained relatively light, at approximately 200 patients per day, for the next several days. Both DMATs WA-1 and OR-2 were released from duty at the airport on Friday, September 9.

Discussion

During the peak 3 days of air transport, 2700 patients were flown from New Orleans International Airport to hospitals; this represents the largest evacuation by air in history. DMAT teams are trained to work with the military in the context of the National Disaster Medical System patient movement system. The collaboration and integration of the DMATs and Joint Task Force Katrina to move patients to definitive care was one of the key successes of the airport mission. The service and dedication of the DMAT members cannot be overemphasized. The DMATs provided remarkable and life-saving care under the most challenging circumstances.

However, it must be acknowledged that inadequate medical staffing and organizational lacunae led to suboptimal care for many of the victims of the hurricane who were cared for at the New Orleans International Airport. Approximately 25 people died at the airport between August 31 and September 3. Louisiana's secretary of Health and Hospitals, Fred Cerise, who was in charge of evacuating Charity and University Hospitals, stated, "People have died and will die because there's not enough resources."[16] The number who died en route to hospitals after evacuation from the airport, and the potential reduction in this toll had there been adequate medical staffing, is unknown.

There has been much in the popular press regarding the deficiencies of the federal response to Hurricane Katrina. Deficiencies in organization at the federal level adversely affected the DMATs' ability to provide medical care at the New Orleans International Airport.

Despite the general tumult, some organizations stand out for their efficiency and professionalism, and deserve commendation for their efforts. The US military—primarily air force and army active duty, guard, and Reserve—transported patients and kept order; medical operations would have been impossible without their presence. The US Forest Service provided hot meals, handwashing stations, showers, and laundry services for airport personnel, all of which had a markedly positive effect on staff morale. Volunteers with PRC Compassion, a faith-based charitable organization comprised of a network of churches and ministries,[17] cleaned the airport and undertook basic nursing care of patients and evacuees, endearing themselves to patients and medical staff alike. A significant portion of New Orleans medical staff remained at hospitals and other healthcare facilities despite loss of electricity and all other routine services. DMAT leaders deserve credit for maintaining professionalism during adverse conditions, and providing daily briefings to team members; DMAT team members worked extremely long hours with diligence and compassion, maintained their focus on patient care, and did not complain.

Conclusion

The medical response to Hurricane Katrina was hampered by organizational lapses at local, state, and federal levels. Since the terrorist attacks of September 11, 2001, the US has spent tens of billions of dollars on disaster prevention, preparedness, and response. While the primary focus of these monies has been terrorism, much of these capabilities should have been transferable to a non-terrorist event such as a hurricane. To the extent that the response to Hurricane Katrina served as a test run of the United States' ability to cope with a wide-scale disaster, it is apparent that emergency preparedness in the US remains inadequate (Tables 1 and 2).

Table 1 Acronyms.

AFSOC	Air Force Special Operations Command
DMAT	Disaster Medical Assistance Team
FEMA	Federal Emergency Management Agency
NDMS	National Disaster Medical System
USFS	United States Forest Service
VA	Veteran's Administration
VMAT	Veterinary Medical Assistance Team

Table 2 Chronology of Hurricane Katrina.

(Central daylight-saving time)

Friday, August 26, 2005
11 p.m. Hurricane Katrina landfall near New Orleans forecast.

Saturday, August 27
4 a.m. Katrina is a Category 3 hurricane, sustained winds of 179–209 km/h.

10:24 a.m. Louisiana Governor Blanco asks President George W. Bush to declare a federal state of emergency for Louisiana. Bush later issues declaration giving the Department of Homeland Security and FEMA authority to coordinate "all disaster relief efforts" and free up federal money for Louisiana.

Table 2. (*continued*)

1:30 p.m. Governor Blanco and New Orleans Mayor Ray Nagin strongly urge New Orleans residents to leave, particularly those living in low-lying areas such as the ninth Ward.

5 p.m. Mayor Nagin declares state of emergency and voluntary evacuation of New Orleans.

6 p.m. National Weather Service predicts a 45 percent chance that a Category 4 or 5 hurricane will hit New Orleans.

Sunday August 28
10 a.m. US National Weather Service predicts catastrophic damage to New Orleans. Mass evacuations begin later that morning. Over 10,000 people enter the Superdome for the night at the urging of New Orleans Mayor C. Ray Nagin, who calls the Superdome a ''shelter of last resort.'' The number of evacuees inside the Superdome eventually reached over 30,000 people.[18]

4:30 p.m. The last passenger jet departs from New Orleans International Airport; this will be the last scheduled passenger flight to leave the airport for 16 days.[19]

Monday, August 29
5:35 a.m. The hurricane makes landfall at the Louisiana–Mississippi border of the Gulf Coast. Accompanying the hurricane is a 8.8 m surge of ocean water, the highest ever recorded.

8 a.m. Hurricane Katrina passes 32 km to the east of New Orleans.
11 a.m. Floodwall of the Industrial Canal breaks open in two places, flooding the ninth Ward with 1–3 m of water. Thousands of residents climb to their rooftops.

Tuesday August 30
1:30 a.m. 17th Street Canal barriers along two blocks fail, flooding 80% of New Orleans. The local pump station fails. A 100 m section of the floodwall lining the London Avenue Canal fails, worsening the flooding.

9 a.m. First helicopter arrives at airport with evacuees from rooftops and hospitals.

Wednesday, August 31
1 a.m. Initial 3 DMATs arrive at Louis Armstrong New Orleans International Airport.
Over 25,000 evacuees are in the Superdome, which is 1 m deep in water.
Floodwaters continue to rise.
Efforts to sandbag the failed 17th Street Canal barriers begins.[20]

Thursday September 1
3 p.m. DMAT OR-2 arrives at Louis Armstrong New Orleans International Airport.

Friday, September 2–Saturday September 3
Medical staffing remains inadequate to address even basic nursing care.

Table 2. (*continued*)

Sunday, September 4
Increasing staff and a lessened flow of incoming patients allowed transport of surviving occupants of the hospice to hospitals; thereafter no patients were designated to receive hospice care only.

Friday, September 9
DMATs OR-2 and WA-1 leave the airport and fly home the next day.

Acknowledgements

The authors are indebted to Joel McNamara for his research contributions and his after-action report (written with two of the authors [Jui and Miller]) on DMAT OR-2's deployment to the New Orleans International Airport.

References

1. Associated Press article, December 20, 2006: 〈http://www.usatoday.com/weather/hurricane/2005-12-20-katrina-strength_x.htm〉 [accessed 12/26/05].
2. FEMA website: 〈http://teams.fema.gov/dmat/〉 [accessed 9/25/05].
3. Oregon Disaster Medical Team website 〈http://www.odmt.org/links.html〉 [accessed 9/25/05].
4. Internet site for Louis Armstrong New Orleans Intenational Airport: 〈http://www.flymsy.com/〉 [accessed 12/15/05].
5. airnav.com website: 〈http://www.airnav.com/airport/KMSY〉 [accessed 12/15/05].
6. Briggs SM, Leong M. Classical concepts in disaster medical response. In: Leaning J, Briggs SM, Chen LC, editors. *Humanitarian crises*. Cambridge: Harvard University Press; 1999. p. 69–79.
7. Vankawala, Hemant reported to Josh Fischman, US News and World Report website: 〈http://www.usnews.com/usnews/health/articles/050910/10emergency.htm〉 posted 9/10/05 [accessed 9/25/05].
8. 〈http://www.afa.org/magazine/Nov2005/1105edge.asp〉 [accessed 12/15/05].
9. Sively, Bruce 〈http://www.flymsy.com/〉 [accessed 12/15/05].
10. Air Force Medical Service website: 〈http://www.airforcemedicine.afms.mil/sg_newswire/hurricane_katrina/AirmenSavingLivesLeadStory.htm〉 [accessed 12/15/05].
11. Federal Aviation and Aeronautics (FAA) Fact Sheet: 〈http://www.faa.gov/news/news_story.cfm?type=fact_sheet&year=2005&date=091505〉 [accessed 12/21/05].
12. Journal of the Air Force Association website: 〈http://www.afa.org/magazine/Nov2005/1105edge.asp〉 [accessed 12/15/05].
13. Journal of the Air Force Association website: 〈http://www.afa.org/magazine/Nov2005/1105edge.asp〉 [accessed 12/15/05] (Jo of AF Assoc. Website).
14. Federal Aviation and Aeronautics (FAA) Fact Sheet: 〈http://www.faa.gov/news/news_story.cfm?type=fact_sheet&year=2005&date=091505〉 [accessed 12/21/05].
15. Vankawala, Hemant reported to Josh Fischman, US News and World Report website: 〈http://www.usnews.com/usnews/health/articles/050910/10emergency.htm〉 posted 9/10/05 [accessed 9/25/05].
16. Childress, Sarah: Critical Condition. *Newsweek*, September 12, 2005, p. 51.

17. PRC Compassion web site: 〈http://www.prccompassion.org/〉. [accessed 10/1/05].
18. House Select Katrina Response Investigation Committee 〈http://www.ngaus.org/ngaus/files/ccLibraryFiles/Filename/000000000656/House%20Select%20Katrina%20Response%20Investigation%20Committee%20Holds%20Hearing%20on%20Hurricane%20Katrina%20Preparedness%20and%20Response.pdf〉 [accessed 12/15/05].
19. Hurricane Katrina from the Airport's Point of View. 〈http://www.flymsy.com/Katrinastory.htm〉 [accessed 9/25/05].
20. Thomas, Evan: The lost city. *Newsweek*, September 12, 2005, pp. 46B-C.